Signal Rescue Lab

Instructor answer key | Hamming(7,4) | Even parity

Use this key to check reasoning, not only final codewords. A correct syndrome without visible parity counts is not yet independent mastery.

1A The impossible repair

The missing bit cannot be determined from the damaged string alone. Both 0 and 1 are possible unless the sender adds an agreed rule or redundancy.

1B Even-parity sprint

data	check bit	total 1s	audit
101	**0**	2	even
111	**1**	4	even
000	**0**	0	even
0101	**0**	2	even

1C Address detective

position	1	2	3	4	5	6	7
check 1	YES		YES		YES		YES
check 2		YES	YES			YES	YES
check 4				YES	YES	YES	YES

Addresses: position 3 = 1+2; position 5 = 1+4; position 6 = 2+4; position 7 = 1+2+4.

Facilitator move: students should say which questions contain the position before adding the labels.

Seven-slot machine and warm-ups

Answer key page 2

2A Roles and check groups

1	2	3	4	5	6	7
p1	p2	d1	p4	d2	d3	d4
check 1	check 2	data 1	check 4	data 2	data 3	data 4

position	1	2	3	4	5	6	7
check 1	YES		YES		YES		YES
check 2		YES	YES			YES	YES
check 4				YES	YES	YES	YES

2B Training wheels: message 1 0 1

1	2	3	4	5	6	7
1	0	1	1	0	1	0
check 1	check 2	data 1	check 4	data 2	data 3	fixed 0

check 1: 1+1+0+0=2	check 2: 0+1+1+0=2	check 4: 1+0+1+0=2

Completed word: 1011010.

2C Training wheels: message 0 1 1

1	2	3	4	5	6	7
1	1	0	0	1	1	0
check 1	check 2	data 1	check 4	data 2	data 3	fixed 0

Completed word: 1100110. Each parity group contains two 1s.

Terminology: the warm-up fixes position 7 at zero. It is a restricted subcode of Hamming(7,4), not the standard Hamming code.

Full Hamming(7,4) encoding

Answer key page 3

3A Four-move routine

1 label positions 1-7; 2 place data in 3,5,6,7; 3 choose p1,p2,p4 for even parity; 4 verify all three checks.

3B Message 1 0 1 1

1	2	3	4	5	6	7
0	**1**	**1**	**0**	**0**	**1**	**1**
check 1	check 2	data 1	check 4	data 2	data 3	data 4

check 1: 0+1+0+1=2	check 2: 1+1+1+1=4	check 4: 0+0+1+1=2

Codeword: 0110011. Check bits: p1=0, p2=1, p4=0.

3C Speed round

message	codeword	check bits p1,p2,p4
0000	**0000000**	0,0,0
1111	**1111111**	1,1,1
0101	**0100101**	0,1,0
1100	**0111100**	0,1,1

Common error: placing message bits consecutively in positions 1-4. Require the role row before any parity arithmetic.

Repair and decode

Answer key page 4

4A Repair 0 1 1 0 0 0 1

1	2	3	4	5	6	7
0	1	1	0	0	0	1
check 1	check 2	data 1	check 4	data 2	data 3	data 4

check	positions	1s	result	label
1	1,3,5,7	2	pass	0
2	2,3,6,7	3	fail	2
4	4,5,6,7	1	fail	4

Syndrome = 2+4 = 6. Flip position 6. Corrected word = 0110011. Recovered message = 1011.

4B Packet repair race

packet	received	failed	corrected	message
A	0101101	4	0100101	0101
B	0111101	1+2+4=7	0111100	1100

No-error case: if all checks pass, syndrome 0 means leave the codeword unchanged. Under the single-error promise, any syndrome 1-7 tells which bit to flip.

Exchange, certification, and bonus

Answer key page 5

5A-5B Partner exchange

Answers vary. A valid sender word has even parity in all three groups. The receiver's syndrome must equal the sender's secretly flipped position, and extraction after repair must reproduce the sender's four-bit message.

Extension: ask early finishers to flip a check bit (1,2,4) and compare with flipping a message bit (3,5,6,7). Both are corrected; only a data-bit error changes the extracted message before repair.

6A Solo certification

part	result	checks / syndrome	final message
A	**0011001**	p1=0, p2=0, p4=1	1001
B	**error at 5; corrected 0011001**	failed 1+4	1001

6B Extended Hamming(8,4) decision table

syndrome	overall parity	receiver action
0	even	**no error**
nonzero	odd	**correct one error in positions 1-7**
0	odd	**flip overall parity bit 8**
nonzero	even	**detect two errors; do not correct**

Session mastery criterion: the student can encode and repair the solo certification without a partner, while showing the three check calculations.

Source: R. W. Hamming, 'Error Detecting and Error Correcting Codes,' Bell System Technical Journal 29(2), 1950, pp. 147-160. https://doi.org/10.1002/j.1538-7305.1950.tb00463.x